SPE-227219-MS

Nanocomposite Electrocatalysts for Efficient Urea Electrolysis in Industrial and Medical Waste Treatment

Hissah A. Alqahtani, Al Yamamah University, Al Khobar, Saudi Arabia; Sheikha Lardhi, King Fahad University of Petroleum and Minerals, Al Khobar, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

In this study, NiFe-LDH catalysts were fabricated on stainless steel mesh (SSM) via electrochemical deposition at varying deposition times. Electrochemical evaluations revealed that among the prepared samples, the NiFe-LDH electrode deposited for 1000 seconds exhibited the highest performance and stability for the urea oxidation reaction (UOR) and the NiCo-LDH electrode exhibited the highest performance for hydrogen evolution reaction (HER) in 1.0 M KOH with 0.5 M urea, as confirmed by linear sweep voltammetry (LSV). The excellent catalytic activity of the NiFe-LDH/SSM/1000s electrode underscores its potential as an efficient non-noble metal catalyst. In addition to electro-chemical testing, structural and surface characterizations were carried out using X-ray diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR), while density functional theory (DFT) calculations were performed to gain insight into the electronic structure and catalytic activity of the NiFe-LDH system.

Introduction

The increasing depletion of fossil fuel reserves has intensified interest in renewable energy alternatives. Electrochemical technologies, particularly the hydrogen evolution reaction (HER), oxygen evolution reaction (OER), and urea oxidation reaction (UOR), are gaining traction due to their promise for clean and efficient energy generation (Li, Y. et al. 2021). Among these, urea electrolysis stands out because it operates at a significantly lower potential (0.37 V) related to traditional water splitting (1.23V), enabling considerable energy savings—up to 93%. Urea is not only rich in hydrogen (6.67 wt%) but also composed of carbon, hydrogen, nitrogen, and oxygen, making it an attractive anodic fuel in electrochemical applications such as fuel cells (Kong, D. et al. 2018).

Urea is widely available from various sources: it naturally occurs in human and animal urine (2.5 wt %, ~ 0.33 mol L^{-1}), is present in wastewater, and is extensively used in industries for fertilizer production, melamine synthesis, and diesel exhaust treatment (Chen, S. et al. 2016). Consequently, urea offers diverse applications in hydrogen production, pollutant degradation, and direct urea fuel cells (Sun, X. J. et al. 2020). However, improper disposal of urea can lead to environmental challenges. Its hydrolysis can release ammonia, contributing to environmental pollution and acid rain (L.M. Le Leuch, T.J. Bandosz

2007, Sung Dae Yim, Soo Jean Kim, Joon Hyun Baik, In−Sik Nam, Young Sun Mok, Jong-Hwan Lee, Byong K. Cho, Se H. Oh 2004). Oxidation of ammonia further leads to hazardous compounds such as nitrites, nitrates, and nitrogen oxides (NO) which contaminate water and soil. Conventional urea treatment methods include hydrolysis, adsorption, chemical oxidation, and biological degradation are often costly and energy-intensive, requiring high temperatures that can deactivate biological enzymes. In contrast, electrochemical UOR provides an efficient route, enabling large-scale operations under mild conditions with stable production of carbon dioxide and nitrogen in alkaline media. Thus, integrating UOR in electrolysis not only improves hydrogen production efficiency but also aids in wastewater treatment.

However, the slow kinetics of both reactions in urea and water electrolysis require the use of electrocatalysts that accelerate the process and enhance energy productivity. However, the trivial noble catalysts, for example, IrO_2 RuO_2, and Pt/C exhibit excellent catalytic activity, their shortage and lower cost effectiveness hinder large-scale application.(Yan, L. et al. 2019) Consequently, there is increasing concern in developing cost-effective and effective non-noble metal electrocatalysts, including phosphides, macrocycles, nitrides, transition metal compounds, sulfides and oxides. Between these, transition metal phosphides, for example, Ni_2P,(Wang, X. D. et al. 2017) CoP,(Liao, H. B. et al. 2018) MoP,(Teng, Y. et al. 2017) and FeP(Yan, Y. et al. 2018) have shown substantial attention due to their favorable wettability and conductivity. Notably, molybdenum phosphide has appeared as a favorable candidate based on its large scale availability, catalytic activity, stability, and applicability across a broad pH range.(Huang, C. et al. 2018) The catalytic efficiency of molybdenum phosphide still lags behind that of noble metal catalysts, ongoing research has led to notable advancements. One effective strategy to enhance molybdenum phosphide performance is to create composite catalysts by integrating it with other materials.

Nevertheless, the sluggish kinetics of the UOR at the anode necessitates the development of efficient, low-cost catalysts. While noble metals have been extensively studied, their limited availability and high cost have spurred interest in more affordable options. Transition metals such as Fe, Co, Ni, Mn, and Mo, along with their compounds (phosphides, sulfides, oxides, hydroxides, nitrides and selenides), have shown promise in both water splitting and UOR applications applications (Chun Yin, Fulin Yang, Shuli Wang, Ligang Feng 2023, Li, J. X. et al. 2022, Li, J. X. et al. 2024, Shenyi Song, Xingyu Huang, Yun Yang, Ligang Feng 2024, Yuanjian Li, Huanwen Wang, Rui Wang, Beibei He, Yansheng Gong 2018). Among these, layered double hydroxides (LDHs) and their derivatives are particularly attractive because of their richness, cost-effectiveness, and catalytic potential (Chen, Gao-Feng et al. 2016, Hu, C. L. et al. 2019, Zhang, H. J. et al. 2020). In their structure, divalent cations (such as Fe^{2+}, Ni^{2+}, Co^{2+},) are partly substituted by trivalent cations (such as Fe^{3+}, Al^{3+}, Mn^{3+}), resulting in a positively charged framework balanced by various interlayer anions (such as NO_3^-, CO_3^{2-}, Cl^-, SO_4^{2-}). These hydroxide layers contain hydroxyl groups that can interact with interlayer species via hydrogen bonding. The general formula for LDHs is $\left[M_{1-x}^{2+}M_x^{3+}\left(OH\right)_2\right]^{x+}\left[A_{x/m}^{m-}.nH_2O\right]$, where M^{2+} and M^{3+} symbolize divalent and trivalent metal cations, respectively, and A_m^- is the interlayer anion (Gong, Y. M. et al. 2022, Sahoo, D. P. et al. 2022). This tunable structure allows for adjustable metal and anion compositions and exfoliation into ultrathin nanosheets, which enhances their catalytic properties (Yang, Mengya et al. 2022).

LDHs are well-suited for UOR due to their high surface area, structural tunability, and favorable redox behavior. The synergistic effects among transition metals (e.g., Ni, Fe, Co) further enhance their catalytic efficiency (Wang, Y. et al. 2018). Additionally, the interlayer space in LDHs can be engineered to facilitate urea and intermediate diffusion, thereby improving reaction kinetics.[33] Tailoring their electronic structure through doping, surface modification, and alloying has led to substantial progress in enhancing UOR performance (Li, P. et al. 2023). Despite these advancements, a deeper understanding of the mechanistic role of LDH-based catalysts is still needed to fully optimize their potential for industrial-scale applications.

These materials display unique surface characteristics such as swelling ability, biocompatibility, high thermochemical stability, and tunable surface chemistry—primarily due to the presence of oxo-bridged and hydroxyl functional groups. Furthermore, doping LDHs with various metal ions enables precise control over their electronic structure and morphology, making them highly adaptable for diverse applications. (Mengya Yang, Rong Zhao, Yi Liu, Hua Lin 2022) However, a major drawback of LDH flakes is their tendency to stack or agglomerate, which limits their performance and broader applicability.(Wang, Z. et al. 2022) This has led to a surge in research focused on developing innovative fabrication strategies for multi-layered, multifunctional LDH-based composites, particularly those supported by carbon and its derivatives, to prevent aggregation and optimize catalytic performance(Lotfi, N. et al. 2019, Pirkarami, A. et al. 2019) The most efficient LDHs such as LDH,(Song, F. et al. 2014) NiFe-LDH,(Fan, K. et al. 2016, Tang, Y. H. et al. 2020) NiCo-LDH,(Cai, X. Q. et al. 2015, Fan, R. L. et al. 2020, Qin, K. Q. et al. 2018) and CoMn offer exceptional electrical conductivity and structural advantages. NiCo-LDH exhibits strong chemisorption for water molecules, effectively adsorbing hydroxyl groups and facilitating water separation.(Yang, Y. Q. et al. 2019) Despite these advantages, NiCo-LDH frequently suffers from small specific surface area due to hydrogen bonding interactions and particle aggregation within its layered structure. To overcome this limitation, it is crucial to employ suitable supporting substrates that enhance surface area and minimize the resistance of electron and charge transfer.(Cai, Z. Y. et al. 2019) By the way, nickel foam with its 3D architecture, having high electrical conductivity, helps in promoting efficient electron transport of electron, improved electrolyte penetration with better contact of active sites.(Long, X. et al. 2016)

In recent years, various transition-metal-based oxides and hydroxides (e.g., Ni, Co, Mo, Fe) have been developed as electrocatalysts for the HER, OER and UOR due to their excess and economy (Babar, P., Lokhande, A., Karade, V., Pawar, B., et al. 2019, Wang, C. et al. 2020, Yan, W. et al. 2012). Whereas Fe and Ni-based hydroxides demonstrate impressive electrocatalytic performance for both UOR and OER in alkaline media, sometimes surpassing valuable metals like Pt, Ru and Ir.(Jadhav, H. S. et al. 2020, Yang, W. L., Yang, X. P., Hou, C. M., et al. 2019, Yang, W. L., Yang, X. P., Li, B. J., et al. 2019) However, Fe and Ni-based electrocatalysts typically deteriorate due to poor HER performance because of the slow kinetics of dissociation of water (the Volmer step).(Guo, Z. G. et al. 2019, Li, Pengsong et al. 2018, Ye, W. et al. 2018) Additionally, bimetallic NiFe hydroxides tend to show enhanced catalytic activity compared to their monometallic counterparts, suggesting a synergistic phenomenon between Fe and Ni that optimizes the bond strength between metal-oxygen, thereby accelerating catalytic kinetics(Babar, P., Lokhande, A., Karade, V., Lee, I. J., et al. 2019, Gao, R. et al. 2017). These challenges drive the need to discover a cutting-edge NiFe OH-based electrocatalysts for efficient, low-energy HER and UOR in LDH structure. In another study, the authors report the synthesis of nanosheets of ultrathin NiFeRh-LDH using ethylene glycol ($C_2H_6O_2$) growth process. The NiFeRh-LDH sample demonstrates exceptional OER and HER activity, likely because of shifts in d-band center with presence of oxygen-vacancies. Notably, they developed cell electrolyzer by NiFeRh-LDH functioning as cathode and anode, achieving current density values of (10, 100, 200)mA/cm^2 at (1.455, 1.565, 1.62)V. Additionally, NiFeRh-LDH catalyst shows exceptional urea oxidation reaction, showing 1.346V to drive 10mA/cm^2 in 1M potassium hydroxide with 0.33M urea. The urea-mediated electrolysis cell designed with NiFeRh-LDH operates at a remarkably low potential (1.35V to 10 mAc/m^2), 105mV lower than the urea-free complement. Whereas, the density functional theory further confirm that introducing Rh into NiFe-LDH structure optimizes the adsorption free energy of urea on NiFeRh-LDH. These outcomes highlight the potential of multifunctional catalysts for the generation of low green energy and the removal of urea from wastewater (Sun, H. C. A. et al. 2021). Moreiver, Chen et al., reported a novel 3D hierarchical CoO–Co_4N@NiFe-LDH/NF electrocatalyst by with hydrothermal, nitridation, and electrodeposition process. The integration of NiFe-LDH with CoO–Co_4N enhances transfer of charge and exposes abundant the active sites, significantly boosting catalytic performance. The resulting catalyst demonstrates excellent activity for the HER, OER and UOR in alkaline media, along with remarkable stability. Particularly, the catalyst attains current density of 10 mA/cm^2 at just 66 mV for 231mV for OER and HER. When used as cathode and anode, it exhibits efficient overall water splitting with a cell voltage of

1.529V. Notably, substituting OER with UOR at the anode reduces the required cell voltage to only 1.39V to sustain $10mA/cm^2$, which is 0.136V lower as compared to conventional water electrolysis—highlighting its potential for energy-saving electrochemical applications (Chen, Baojin et al. 2021).

In this work, we report the fabrication of NiFe-LDH catalysts on stainless steel mesh (SSM) via electrochemical deposition and their application for urea oxidation reaction. The novelty of this study lies in the integration of non-noble, earth-abundant NiFe-LDHs for simultaneous urea degradation and hydrogen evolution under alkaline conditions. The structural and electronic properties of the catalysts were thoroughly investigated using X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), and density functional theory (DFT) calculations.

Methodology

Chemicals and Materials

All reagents used in this study were of analytical grade and used as received without further purification. The following chemicals were utilized for the synthesis of layered double hydroxide (LDH) composites: nickel(II) sulfate hexahydrate ($NiSO^4·6H_2O$, ≥98%), copper(II) sulfate pentahydrate ($CuSO^4·5H_2O$, ≥99%), cadmium(II) sulfate octahydrate ($CdSO^4·8H_2O$, ≥98%), cobalt(II) sulfate heptahydrate ($CoSO^4·7H_2O$, ≥99%), iron(III) chloride ($FeCl^3$, ≥98%), sodium nitrate ($NaNO^3$, ≥99%), urea ($CO(NH_2)_2$, ≥99.5%), potassium hydroxide (KOH, ≥85%), hydrochloric acid (HCl, 37%), absolute ethanol (C_2H^5OH, 99.9%), and deionized (DI) water (resistivity ≥18.2 MΩ·cm). Stainless steel mesh (SSM) was employed as the substrate for electrode preparation.

Pretreatment of Stainless Steel Mesh Electrodes

The stainless steel mesh was cut into rectangular strips with dimensions of 1.0 cm × 2.0 cm. To remove native oxide layers and surface contaminants, the samples were immersed in 0.5 M HCl solution for 10 minutes. This acid treatment facilitates surface activation and improves the adhesion of the deposited LDH layer by etching and roughening the surface. Following acid treatment, the electrodes were sequentially sonicated in absolute ethanol and DI water for 15 minutes each to eliminate organic residues and soluble impurities, respectively. The cleaned electrodes were then air-dried at room temperature and stored in a clean desiccator prior to use.

Electrodeposition of LDH Composites

The deposition of NiM-LDH composites (M = Cu^{2+}, Cd^{2+}, Co^{2+}, Fe^{3+}) was carried out via a potentiostatic method using chronoamperometry (CA) in a conventional three-electrode electrochemical cell. The electrodeposition was conducted at a constant potential between –1.0 V and –1.2 V versus the saturated calomel electrode (SCE) for 1000 s. The electrolyte for each deposition contained 0.05 M $NiSO^4·6H_2O$ and 0.025 M of one of the secondary metal precursors ($CuSO^4·5H_2O$, $CdSO^4·8H_2O$, $CoSO^4·7H_2O$, or $FeCl^3$), with 0.1 M $NaNO^3$ added as a supporting electrolyte to enhance ionic conductivity and facilitate uniform film formation.

During the deposition, the SSM served as the working electrode (WE), a platinum wire as the counter electrode (CE), and SCE as the reference electrode (RE). The effective exposed surface area of the working electrode was confined to 1.0 cm^2. After deposition, the electrodes were rinsed with DI water and dried at room temperature.

Electrochemical Measurements

The electrocatalytic performance of the fabricated electrodes was evaluated using electrochemical impedance spectroscopy (EIS) and linear sweep voltammetry (LSV) in a urea-containing alkaline solution. All tests were performed in a three-electrode configuration, with the modified SSM as the working electrode, platinum as the counter electrode, and SCE as the reference electrode. The electrolyte consisted of 1.0 M KOH and 0.5 M urea and was maintained at room temperature.

EIS measurements were carried out over a frequency range of 0.1 Hz to 100 kHz with an AC amplitude of 10 mV. LSV was performed at a scan rate of 5 mV s^{-1} to assess the activity of the electrodes for both the urea oxidation reaction (UOR) and hydrogen evolution reaction (HER). The potential window for UOR was set from –0.2 V to +0.8 V vs. SCE, while that for HER ranged from –1.0 V to 0.0 V vs. SCE.

Based on the electrochemical analysis, the LDH composition exhibiting the best bifunctional catalytic performance was further optimized by varying the electrodeposition time (500 s and 1500 s), and the resulting electrodes were evaluated under identical electrochemical conditions for comparison.

Characterization

The structural, morphological, and compositional properties of the synthesized LDH/SSM samples were characterized using a combination of X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), and scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM/EDX). XRD analysis was performed to investigate the crystalline structure and phase composition of the LDH material. The diffraction patterns were recorded using a Cu Kα radiation source (λ = 1.5406 Å), and the data were used to confirm the formation of the layered double hydroxide structure.

FTIR spectroscopy was employed to identify the functional groups and interlayer species present in the LDH. The spectra were recorded in the range of 4000-400 cm^{-1}, revealing characteristic bands corresponding to O-H stretching vibrations, interlayer water, and nitrate anions (NO^{3-}) derived from the use of $NaNO^3$ during synthesis. Metaloxygen stretching vibrations observed in the fingerprint region further confirmed the formation of Ni-O and Fe-O bonds within the hydroxide layers.

The surface morphology and elemental composition of the LDH coatings on stainless steel mesh (SSM) were examined using SEM/EDX. SEM provided high-resolution images of the LDH layer, allowing for evaluation of surface coverage, and uniformity. EDX analysis was used to confirm the elemental distribution of nickel, iron, and oxygen, verifying the successful deposition of the NiFe/LDH composite onto the SSM substrate

Theoretical Model and Method

The chemical formula for NiFe LDH is typically represented as $[Ni^{2+}{}_{1-x}Fe^{3+}{}_{x}(OH)_2]^{x+}[A^{n-}]^{x-}\cdot mH_2O$, where A^{n-} denotes the anions intercalated between the layers (Chen, R. et al. 2019). The structural integrity and stability of NiFe LDH are significantly influenced by the choice of intercalated anions, such as NO^{3-}, which can modulate its electrochemical properties. The crystal structure of NiFe LDH is characterized by the monoclinic space group C 2/m (12), indicating the presence of a two-fold rotational axis and a mirror plane within the unit cell. The cell dimensions are defined by a-axis = 5.3545(24) Å, b-axis = 9.142(4) Å, and c-axis = 7.8296(15) Å, reflecting the non-cubic nature of the structure. Notably, the angles are α = 90.0°, β = 103.99(4)°, and γ = 90.0°, revealing a unique angle between the a and b axes that is characteristic of monoclinic systems. In this arrangement, as shown in Figure 1, Ni^{2+} ions occupy octahedral sites, while Fe^{3+} ions can occupy both octahedral and tetrahedral sites within the layered structure. The coordination of Ni and Fe ions leads to the formation of a brucite-like structure, where hydroxide ions (OH^-) are coordinated with the metal cations. NO^{3-} anions neutralize the formation of positively charged layers in the interlayer space.

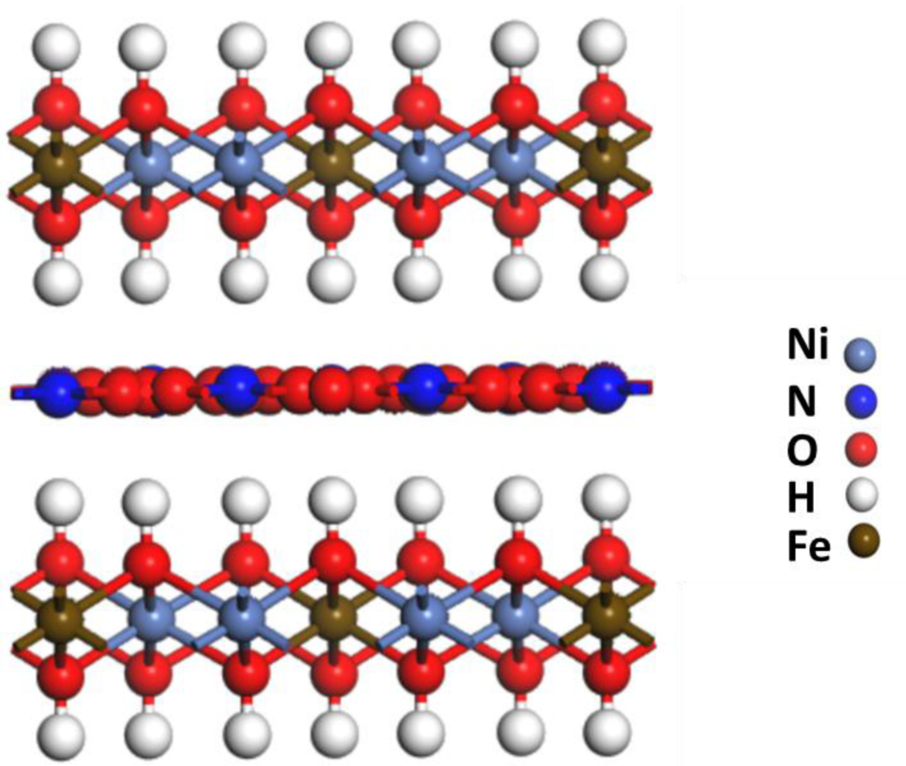

Figure 1—side views of the structure of NiFe-layered double hydroxide (NiFe-LDH) intercalated with nitrate (NO^{3-}) anions. Nickel (Ni) and iron (Fe) occupy octahedral sites within the hydroxide layers, coordinated by oxygen (O) and hydroxyl (OH^-) groups. Nitrate anions are located in the interlayer gallery, oriented between the brucite-like sheets. Color code: Ni (light blue), Fe (brown), O (red), N (dark blue), H (white).

First-principles calculations based on density functional theory (DFT) were carried out using the CASTEP module within the Materials Studio package(Clark, S. J. et al. 2005). The exchange-correlation effects were treated using the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA)(Perdew, J. P. et al. 1996). Ultrasoft pseudopotentials were used to describe the electron–ion interactions. A plane-wave kinetic energy cutoff of 340 eV and a Monkhorst–Pack k-point mesh of 3 × 3 × 3 were used for Brillouin zone integration, based on the anisotropic monoclinic cell dimensions. Geometry optimizations were performed with full relaxation of lattice parameters and atomic positions, using convergence thresholds of 1.0×10^{-5} eV/atom for total energy, 0.03 eV/Å for maximum force, and 0.001 Å for maximum displacement. The electronic band structure and density of states (DOS) were subsequently evaluated based on the optimized geometry.

Results and Discussions

Electrochemical Measurements

Effect of Transition Metal Dopants on UOR and HER Activity of SSM Electrodes. The anodic polarization curves presented in Figure 2 (a) provide critical insights into the electrocatalytic activity of the synthesized NiM-LDH/SSM (M = Fe, Co, Cu, Cd) composites toward the urea oxidation reaction (UOR) in 1.0 M KOH + 0.5 M urea at room temperature. Among all investigated materials, the NiFe-LDH/SSM electrode exhibits the most prominent UOR activity, as evidenced by its highest anodic current density and lowest onset potential compared to other LDH-based electrodes and the bare SSM.

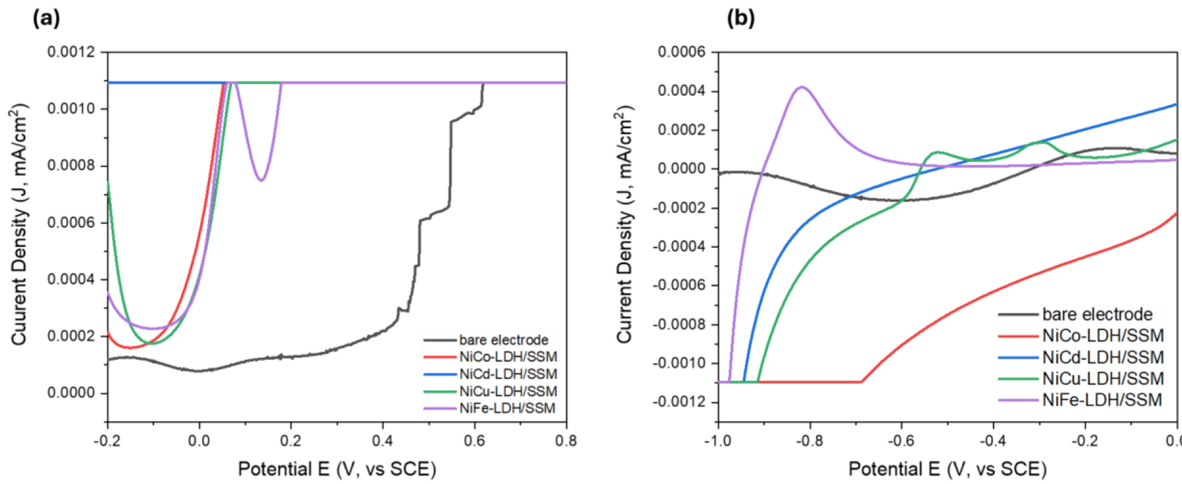

Figure 2—LSV of different LDH composites in 1M KOH + 0.5M urea (a) UOR and (b) HER.

The superior performance of NiFe-LDH may be attributed to the synergistic interaction between Ni^{2+}/Ni^{3+} and Fe^{3+} species. It is well established that the UOR is primarily facilitated by the redox cycling of Ni^{2+}/Ni^{3+} in alkaline medium. However, the incorporation of Fe^{3+} into the LDH structure enhances electron conductivity and promotes the formation of high-valent Ni^{3+}-based active sites (e.g., NiOOH), which are crucial for the oxidative cleavage of urea molecules. The improved conductivity and charge separation also help reduce charge transfer resistance during oxidation. Moreover, the layered structure of LDHs with interlayer spacing and surface hydroxyl groups facilitates urea diffusion and interaction with catalytic sites, further enhancing the electrocatalytic activity (Lv, Q. L. et al. 2025)

The NiCu-LDH/SSM also demonstrated favorable UOR performance, second only to NiFe-LDH. This enhancement can be ascribed to the electronic interaction between Ni and Cu, where Cu^{2+} may participate in synergistically enhancing the Ni active sites or modulating their oxidation state. However, the onset potential is relatively higher, and the current density is lower than NiFe-LDH, indicating moderate activity.

NiCd- and NiCo-LDH/SSM electrodes displayed relatively lower UOR performance. For NiCo-LDH, although cobalt oxides are known to be good oxygen evolution reaction (OER) catalysts, their contribution to urea oxidation appears less effective, possibly due to their limited redox transitions within the potential window relevant for UOR. In contrast, the NiCd-LDH showed minimal enhancement over the bare SSM, likely due to the lack of redoxactive Cd species and a possible blocking effect of Cd^{2+} on Ni active sites (Satheesan, A. K. et al. 2025).

Overall, the order of UOR performance observed from the LSV curves is as follows:

$$\text{NiFe-LDH/SSM} > \text{NiCu-LDH/SSM} > \text{NiCd-LDH/SSM} \approx \text{NiCo-LDH/SSM} > \text{bare SSM}$$

This trend strongly suggests that Fe-based LDHs possess the most favorable properties for promoting urea electro-oxidation in alkaline media, making them highly attractive for use in direct urea fuel cells and urea-rich wastewater treatment.

The cathodic LSV plots presented in Figure 2 (b) reveal the HER activity of the same set of LDH-modified electrodes in 1.0 M KOH and 0.5 M urea. In contrast to the UOR behavior, the NiCo-LDH/SSM electrode demonstrates the best HER performance, evidenced by its lowest onset potential and highest cathodic current density.

The high efficiency of NiCo-LDH toward HER can be attributed to the electronic synergy and bifunctional role of Co^{2+} in water dissociation and hydrogen adsorption. In alkaline HER, the Volmer step ($H_2O + e^- \rightarrow H_{ads} + OH^-$) is typically rate-limiting. Co-based species are known to accelerate this step by facilitating the cleavage of the H-OH bond, leading to a higher availability of adsorbed hydrogen (H_{ads}) on the catalyst surface. When combined with Ni in the LDH framework, the resultant NiCo-LDH offers improved electrical conductivity, enhanced electron mobility, and a higher density of catalytically active sites (Wu, Zhuorun et al. 2025).

Although NiFe-LDH excelled in UOR, it showed moderate HER performance in comparison with NiCo-LDH. This may result from the inherent limitations of Fe^{3+} in the HER process, where its role in proton adsorption and hydrogen desorption is less efficient than that of Co. However, the presence of Fe in NiFe-LDH still contributes to water dissociation and the overall conductivity of the composite, explaining its better performance compared to NiCd- and NiCu-LDH systems.

NiCd-LDH/SSM and NiCu-LDH/SSM demonstrated poorer HER activity. Cd^{2+} is redox-inactive in the HER potential window and may block catalytic sites or disrupt optimal hydrogen adsorption geometry. Similarly, although Cu is a moderately active HER catalyst, it does not form effective active sites within the LDH matrix in alkaline media (Abraham, D. S. et al. 2025). The observed HER activity trend is summarized as follows:

NiCo-LDH/SSM > NiFe-LDH/SSM > NiCd-LDH/SSM > NiCu-LDH/SSM > bare SSM

This behavior highlights the importance of targeted metal selection when designing LDH-based bifunctional catalysts for overall urea electrolysis. While NiFe-LDHs are preferable for anodic oxidation (UOR), NiCo-LDHs offer superior cathodic activity (HER), suggesting a split-electrode system or hybrid composition could potentially deliver optimal overall cell performance (Long, G. et al. 2025).

Electrochemical Impedance Spectroscopy (EIS) is a powerful diagnostic technique to probe charge transfer kinetics and interfacial resistance of electrode materials. The Nyquist plots of the various NiM-LDH/SSM electrodes (M = Co, Fe, Cu, Cd) in 1.0 M KOH and 0.5 M urea, shown in Figure 3 (a) and (b), provide valuable insights into the electrochemical behavior of the catalysts toward both UOR and HER processes.

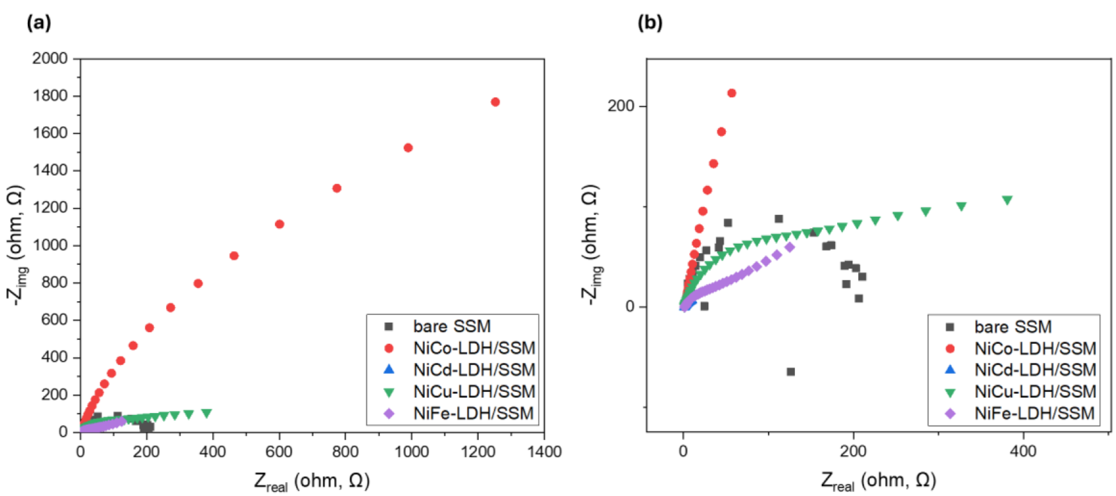

Figure 3—EIS of different LDH compoites in 1M KOH + 0.5M urea, (a) zoom-out (b) zoom-in

The semicircle diameter in the high-frequency region of a Nyquist plot corresponds to the charge transfer resistance (R_{ct}), which is inversely related to the electrocatalytic activity of the electrode. A smaller semicircle indicates a lower R_{ct}, and thus, faster charge transfer and improved catalytic efficiency.

Among all tested materials, the NiCo-LDH/SSM electrode clearly exhibits the smallest semicircle, signifying the lowest R_{ct} and, consequently, the most efficient charge transfer kinetics at the electrode/electrolyte interface. This observation aligns well with the high HER activity demonstrated in the corresponding LSV plots, confirming that Co incorporation into the LDH framework enhances the intrinsic electrical conductivity and facilitates electron transport (Chen, Yun et al. 2025).

Following NiCo-LDH/SSM, NiFe-LDH/SSM and NiCu-LDH/SSM show moderate semicircle sizes, indicating intermediate R_{ct} values. NiFe-LDH is particularly notable, as its relatively low charge transfer resistance correlates with its superior UOR activity. This suggests that Fe incorporation not only improves redox properties and urea adsorption but also helps in maintaining favorable conductivity for urea electro-oxidation. The NiCu-LDH electrode, while having reasonable conductivity, showed slightly higher R_{ct} than NiFe-LDH, which is reflected in its lower catalytic performance (Abdelfattah, Sara A. et al. 2025).

On the other hand, NiCd-LDH/SSM and bare SSM displayed the largest semicircles, indicating poor electron transfer characteristics and high interfacial resistance. This is attributed to the lack of electroactive redox transitions for Cd^{2+} and the absence of a conductive or catalytically active framework in the bare SSM, leading to significantly hindered UOR and HER kinetics.

The inset zoom-in view in Figure 3 (b) enhances the visibility of the lower-resistance electrodes, confirming the superior conductivity of NiCo-LDH and, to a lesser extent, NiFe-LDH. This distinction becomes crucial in electrolysis applications, where efficient charge transport directly impacts overall cell voltage, energy efficiency, and durability.

The EIS results are pivotal in interpreting the trends observed in LSV and CV measurements. For UOR, a lower R_{ct} facilitates rapid electron transfer during the oxidation of urea, enhancing current density and lowering over-potentials. The enhanced conductivity in NiFe-LDH supports its high UOR activity. For HER, fast kinetics of electron delivery to protons (or water molecules in alkaline media) is crucial. The excellent EIS profile of NiCo-LDH supports its top HER performance, confirming the role of low charge transfer resistance in efficient hydrogen evolution.

Effect of Electrodeposition Time on UOR and HER Performance of NiFe-LDH/SSM Electrodes. To evaluate the effect of electrodeposition duration on the electrocatalytic activity of NiFe-LDH/SSM electrodes, linear sweep voltammetry (LSV) was conducted for both urea oxidation reaction (UOR) and hydrogen evolution reaction (HER) using electrodes prepared at deposition times of 500 s, 1000 s, and 1500 s. All measurements were performed in 1.0 M KOH containing 0.5 M urea at room temperature using a three-electrode configuration. The results are illustrated in Figure 4 (a) and (b), corresponding to UOR and HER, respectively.

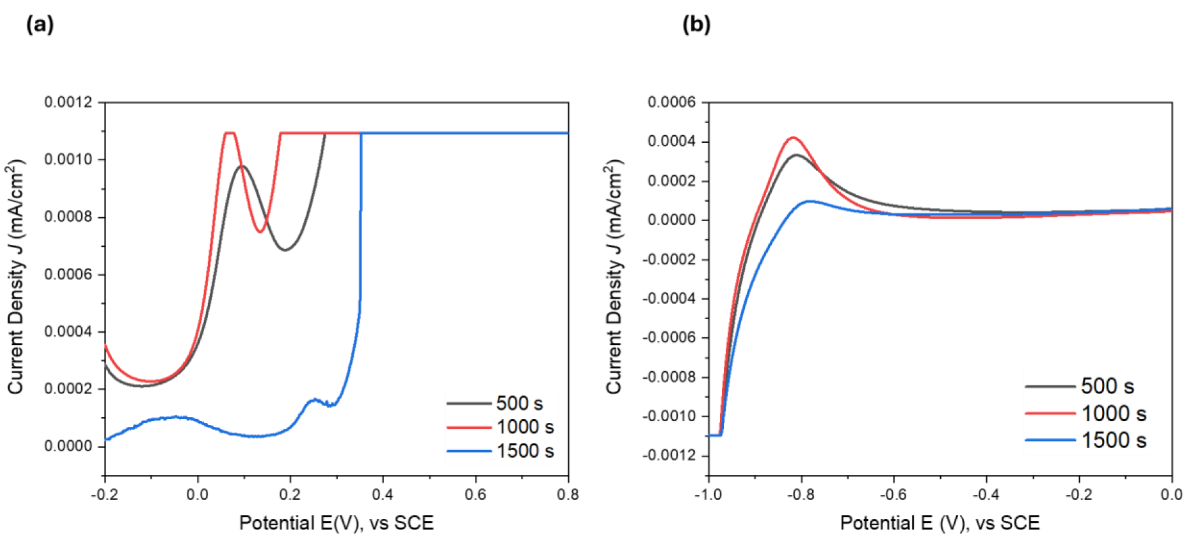

Figure 4—LSV of NiFe-LDH/SSM electrode in 1M KOH + 0.5M urea (a) UOR and (b) HER.

The LSV curves in Figure 4 (a) reveal a distinct trend with respect to deposition time. Among the tested samples, the electrode deposited for 1000 s exhibited the highest anodic current density and the lowest onset potential for UOR, suggesting an optimized surface morphology and electrocatalytically active area for urea oxidation. The onset potential is shifted positively as the deposition time increases from 500 s to 1500 s, indicating a decline in catalytic efficiency at longer deposition times.

The superior performance of the 1000 s sample can be attributed to a balanced growth of the NiFe-LDH layer on the stainless steel mesh (SSM). At this intermediate time, the LDH structure is likely well-developed and uniformly distributed, providing abundant accessible redox-active sites (Ni^{2+}/Ni^{3+} and Fe^{2+}/Fe^{3+} pairs) that can facilitate the six-electron oxidation pathway of urea to N_2, CO_2, and H_2O. Moreover, the relatively lower charge-transfer resistance and improved electron conduction through the interconnected LDH network contribute to the enhanced kinetics.

In contrast, the electrode prepared at 500 s shows moderate UOR activity, likely due to an underdeveloped catalytic layer with fewer active sites and insufficient surface coverage. On the other hand, the sample deposited for 1500 s exhibits a marked decline in performance. This may result from excessive film thickness, which can hinder electron and mass transport, reduce electrochemical surface area (ECSA), and introduce structural instability or flaking. The overgrown LDH film likely limits access to inner active sites and slows down the diffusion of urea molecules to the reactive interface.

The HER LSV curves, displayed in Figure 4 (b), follow a similar trend to UOR, where the 1000 s-deposited NiFe-LDH/SSM electrode demonstrates the best performance, characterized by a higher cathodic current density and a lower onset potential. This optimal behavior highlights the critical role of electrode morphology and conductivity in promoting efficient proton reduction in alkaline medium, where water dissociation (Volmer step) is the rate-limiting stage.

The enhanced HER activity at 1000 s may be attributed to the optimal balance between film thickness, crystallinity, and porosity, ensuring effective electron transport and fast desorption of hydrogen gas bubbles. Additionally, NiFe-LDH is known to exhibit favorable water dissociation properties and synergistic redox interactions between Ni and Fe sites, which are crucial for HER catalysis under alkaline conditions.

The 500 s sample, although thinner and possibly more conductive, offers limited active site density, while the 1500 s film, similar to the UOR case, may suffer from diffusion resistance and blocked catalytic sites due to overloading. These effects collectively result in lower HER activity compared to the 1000 s counterpart.

The combined UOR and HER results suggest that the deposition time plays a decisive role in tuning the electrocatalytic performance of NiFe-LDH-coated SSM electrodes. While a short deposition time (500 s) yields insufficient catalytic material, and an extended period (1500 s) causes over-deposition and poor charge transfer, the intermediate 1000 s duration achieves optimal film morphology, exposing the maximum number of electroactive centers and facilitating charge and mass transport. This is crucial for dual-functional electrodes intended for full-cell urea electrolysis.

Such optimization of deposition time is vital not only for performance enhancement but also for energy efficiency, material utilization, and mechanical durability of the electrode system. The results support the further development of NiFe-LDH nanostructures as promising bifunctional electrocatalysts for integrated wastewater treatment and hydrogen production applications.

FTIR Analysis of NiFe/LDH Composite

The FTIR spectrum of the synthesized NiFe-LDH composite reveals key vibrational bands that are characteristic of layered double hydroxide structures and confirm the successful formation of the NiFe-LDH phase as shown in Figure 5. The broad absorption band observed in the region around 3400 cm^{-1} is attributed to the stretching vibrations of the -OH groups, which are primarily from interlayer water molecules and hydroxyl groups in the brucite-like layers of the LDH. This broadness suggests extensive hydrogen bonding, which is typical in layered hydroxides due to water interactions and interlayer anions.

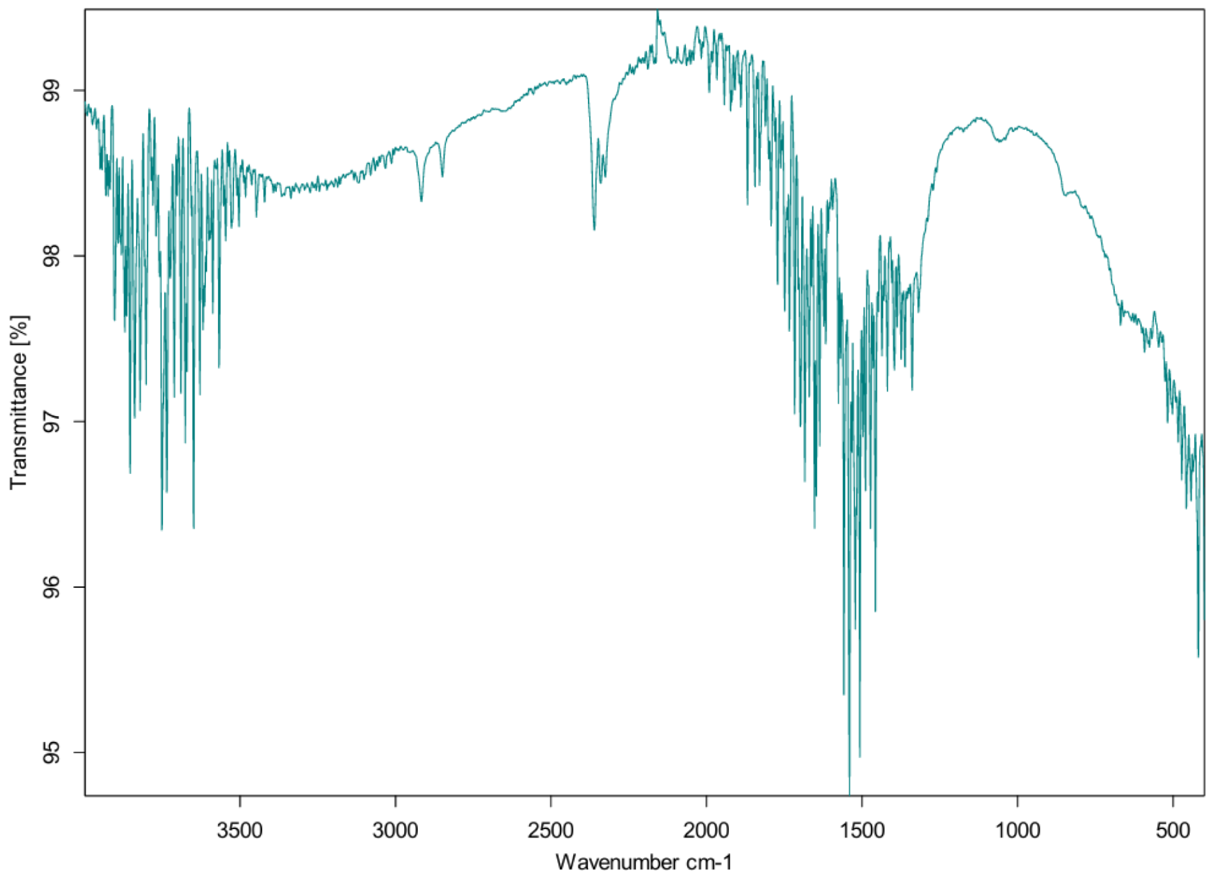

Figure 5—FTIR of NiFe-LDH/SSM electrode

A weak shoulder or bending mode appears around 1600 cm^{-1}, corresponding to the H-O-H bending vibrations of water molecules trapped in the interlayer space. This feature confirms the presence of structural or physically adsorbed water, which plays a crucial role in maintaining the layered structure.

Crucially, a strong and sharp absorption band at ~1384 cm^{-1} is observed, which is characteristic of the symmetric stretching vibration of nitrate anions (NO^{3-}). This confirms the successful intercalation of nitrate ions into the interlayer gallery of the LDH, originating from the use of sodium nitrate ($NaNO^3$) as the supporting electrolyte or precursor during the synthesis process. The presence of nitrate, rather than carbonate, suggests that the synthesis was carried out under conditions that minimized CO_2 uptake, thereby preserving the intended anion composition and enhancing the control over interlayer chemistry (Zeng, Z. Y. et al. 2025).

A strong, complex band observed below 800 cm^{-1}, particularly in the range 400-700 cm^{-1}, corresponds to metal oxygen (M-O) stretching vibrations. These are associated with the Ni-O and Fe-O bonds in the LDH layers, confirming the successful incorporation of both nickel and iron into the hydroxide lattice. The absorption bands in this region are typically assigned to the lattice vibrations of the M(II)-OH and M(III)-OH bonds, essential in constructing the brucite-like layers of the LDH structure.

The absence of peaks related to organic contaminants or precursor residues in the 2800-3000 cm^{-1} region further indicates the purity of the material and successful post-synthesis washing or purification steps.

Overall, the FTIR results support the formation of a well-defined NiFe-LDH structure with interlayer nitrate anions and hydroxyl groups, essential for maintaining the integrity of the layered architecture. These structural features not only confirm the LDH phase but also highlight the potential of the NiFe-LDH composite for applications in electrocatalysis, such as hydrogen evolution reactions (HER) and urea oxidation reactions (UOR), due to the synergistic interactions between nickel and iron centers.

XRD Analysis of LDH Composites

The X-ray diffraction (XRD) patterns in Figure 6 provide structural insight into the bare stainless steel mesh (SSM), NiCo-LDH/SSM, and NiFe-LDH/SSM after 1000 seconds of electrodeposition. The bare SSM exhibits distinct diffraction peaks at 2θ values of approximately 44.5°, 51.8°, and 76.4°, which correspond to the (111), (200), and (220) planes of face-centered cubic (FCC) stainless steel (JCPDS No. 00-047-1417) (Zeng, Z. Y. et al. 2025). These peaks confirm the crystalline metallic substrate and serve as reference for analyzing surface modifications upon LDH deposition.

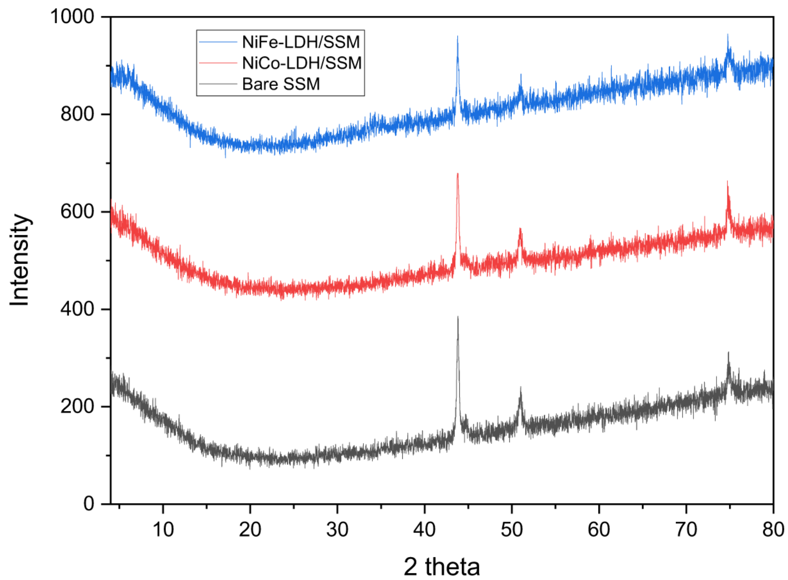

Figure 6—XRD patterns of bare stainless steel mesh (SSM), NiCo-LDH/SSM, and NiFe-LDH/SSM after 1000 seconds of electrodeposition.

Following the electrodeposition of LDH films, the XRD patterns of NiCo-LDH/SSM and NiFe-LDH/SSM reveal broad, low-intensity features in the low-angle region (10°-30°), indicative of the presence of poorly crystalline or amorphous LDH phases. The reduction in overall peak intensity compared to bare SSM suggests the formation of a surface coating that partially obscures the crystalline signal from the underlying steel. In both LDH-modified samples, a subtle shoulder near ~11.5° may correspond to the (003) basal reflection of layered double hydroxides, consistent with the formation of stacked hydroxide layers with intercalated anions.

The absence of sharp, well-defined LDH peaks indicates a low degree of crystallinity, which is commonly observed in electrodeposited LDH films and is beneficial for electrocatalytic applications. Poorly crystalline structures often expose more active edge sites, enhance ion mobility, and increase electrochemically active surface area. These features are desirable in both the hydrogen evolution reaction (HER) and urea oxidation reaction (UOR), where charge transfer and surface reactivity are critical. Notably, the NiFe-LDH/SSM pattern shows a more pronounced suppression of the substrate peaks and higher background scattering, implying a thicker and more uniform coating compared to the NiCo-LDH/SSM sample an observation consistent with SEM morphology.

In summary, the XRD results confirm the successful deposition of LDH layers on SSM, with low crystallinity beneficial for catalytic function. The structural evolution observed across the three samples complements the compositional and morphological findings from the following SEM and EDX analyses, collectively validating the formation of active nanostructured LDH electrocatalysts for water and waste treatment.

SEM/EDX Analysis of LDH Composites

The scanning electron microscopy (SEM) images presented in Figure 7 offer a detailed examination of the surface morphology and microstructural transformation of stainless-steel mesh (SSM) electrodes before and after deposition of layered double hydroxide (LDH) nanocomposite coatings. The figure is structured into three panels: (a) shows the bare SSM electrode, (b) corresponds to NiCo-LDH/SSM after 1000 seconds of electrodeposition, and (c) illustrates NiFe-LDH/SSM under identical deposition conditions. Each panel includes four progressive magnifications (500 μm, 100 μm, 50 μm, and 10 μm), offering multiscale insights into the physical structure and coating characteristics of the electrocatalysts.

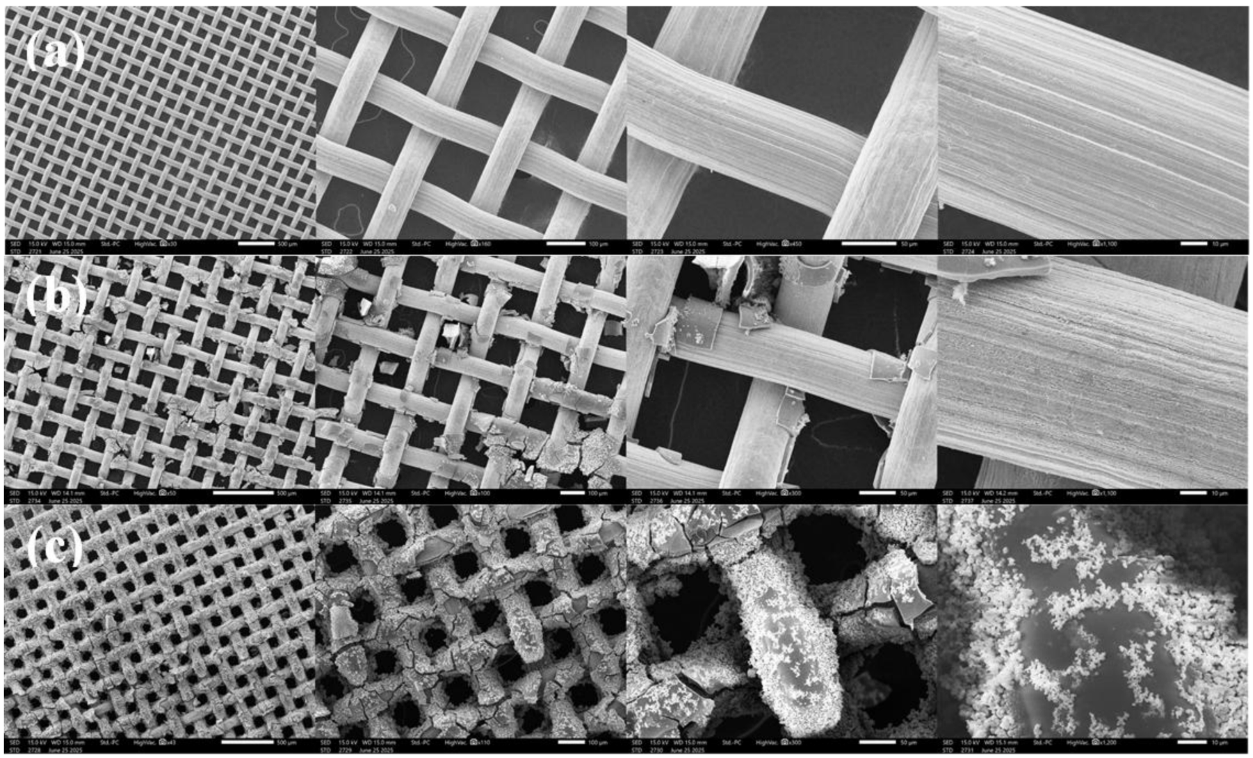

Figure 7—Multiscale SEM micrographs illustrating the morphological evolution of stainless-steel mesh (SSM) electrodes before and after 1000 s electrodeposition of layered double hydroxide (LDH) nanocomposites: (a) pristine SSM, (b) NiCo-LDH/SSM, and (c) NiFe-LDH/SSM. Each row shows four magnifications (left → right: 500 μm, 100 μm, 50 μm, 10 μm scale bars), revealing the transformation from the smooth, bare wire surface to (b) a flake-like NiCo nanosheet network that increases roughness for HER and (c) a dense, flower-like NiFe coating that forms a hierarchical porous film conducive to UOR.

In panel (a), the bare SSM surface reveals a clean, uniform woven structure typical of commercial-grade mesh electrodes. At low magnification (500 μm and 100 μm), the metallic wires appear smooth and tightly interlaced, with well-defined edges and negligible surface roughness. As the magnification increases to 50 μm and 10 μm, the absence of surface deposits confirms the pristine, untreated nature of the steel substrate. This baseline is crucial for evaluating the extent of morphological modification following LDH deposition.

Upon electrodeposition of the NiCo-LDH composite, panel (b) reveals substantial alterations in surface features. At 500 μm and 100 μm, the mesh retains its original framework, but a noticeable textural change is observed on the wire surfaces. A granular coating begins to develop, adhering to both inner and outer surfaces of the mesh strands. Moving to 50 μm and 10 μm, the coating morphology becomes more apparent: thin flakes and nanosheets are distributed across the substrate, forming a partially porous network. The NiCo-LDH layer appears to nucleate in an island-like manner, growing radially and covering available active sites. The observed nanoscale roughness enhances the electrochemically active surface area (ECSA), a key factor in boosting the electrocatalytic activity. These morphological attributes are favorable for promoting hydrogen evolution reaction (HER), as they increase the density of catalytic sites, improve electron transfer kinetics, and facilitate efficient bubble release during gas evolution.

In contrast, the NiFe-LDH/SSM composite (panel c) demonstrates a distinctly different morphology. Even at the lowest magnification (500 µm), the mesh wires appear heavily coated with a denser and more continuous film. As magnification increases to 100 µm and 50 µm, a uniform layer with fractal-like microstructures emerges, displaying significant coverage and texture. At 10 µm, the surface is characterized by flower-like agglomerates and stacked nanosheets, forming a three-dimensional porous architecture. The NiFe-LDH coating is notably thicker and more interconnected than the NiCo counterpart, suggesting stronger nucleation and growth behavior during the electrodeposition process. This densely packed, hierarchical microstructure is particularly advantageous for urea oxidation reaction (UOR), where high surface area, effective mass transport, and abundant active sites are essential. Furthermore, the compact morphology likely improves adhesion to the SSM substrate, ensuring long-term electrochemical stability under alkaline conditions.

Comparing panels (b) and (c), it is evident that the choice of metal ions in the LDH structure critically influences the resulting morphology. NiFe-LDH coatings form denser, more robust films with higher microstructural complexity, which is aligned with the typically superior UOR catalytic activity of NiFe-based materials. Conversely, NiCo-LDH offers a more open, flake-like structure conducive to HER due to improved accessibility of reactants to catalytic centers. These structural distinctions are instrumental in optimizing electrocatalyst design based on the target reaction.

In summary, the SEM analysis provides compelling evidence of the successful deposition of LDH films on SSM substrates and the profound impact of composition on morphology. The bare SSM exhibits a smooth, inactive surface, while NiCo-LDH forms loosely packed nanosheet aggregates suitable for HER. NiFe-LDH, on the other hand, produces a dense, crystalline coating that supports efficient and stable UOR. The structural integrity and functional architecture of these coatings are key to their enhanced performance in urea electrolysis, offering a dual advantage in treating industrial and medical waste streams while producing hydrogen as a clean energy carrier.

The EDX elemental analysis shown in Table 1 presents the atomic percentages of key elements detected on the surfaces of (a) bare stainless-steel mesh (SSM), (b) NiCo-LDH/SSM, and (c) NiFe-LDH/SSM after electrodeposition for 1000 seconds. These results provide a quantitative understanding of the compositional modifications resulting from the formation of Ni-based LDH nanocomposites and further support the morphological transitions observed in the SEM images.

Table 1—EDX atomic percentage analysis of (a) bare stainless steel mesh (SSM), (b) NiCo-LDH/SSM, and (c) NiFe-LDH/SSM after 1000 seconds of electrodeposition.

(a)

Element	C	Cr	Fe	Si	Mn	Ni	O
Atom %	11.48	18.09	61.81	1.05	0.66	6.91	0.001

(b)

Element	C	Cr	Fe	Si	Co	Ni	O
Atom %	8.74	14.10	48.71	0.61	6.16	11.81	9.88

(c)

Element	C	Cr	Fe	Si	Ni	O
Atom %	9.35	0.48	20.98	0.21	5.06	61.10

In sample (a), the bare SSM, the atomic composition is primarily dominated by iron (Fe), which accounts for a substantial portion of the surface, in agreement with the typical composition of stainless steel. Minor percentages of nickel (Ni) and chromium (Cr) are also detected, as expected from the base alloy. The presence of a small amount of oxygen (O) is attributed to the natural formation of surface oxides, while carbon (C) appears due to adsorbed environmental contaminants or a native hydrocarbon layer. The absence of any extrinsic elements confirms the clean, uncoated nature of the electrode, consistent with the smooth, unaltered wire surface observed in the SEM images at all magnifications.

Upon electrodeposition of the NiCo-LDH layer, sample (b) shows a notable decrease in Fe atomic %, reflecting the formation of a surface coating that masks the steel substrate. The elemental profile reveals the introduction of significant amounts of nickel (Ni) and cobalt (Co), verifying the successful incorporation of both transition metals into the LDH framework. Oxygen content increases markedly, which is a strong indicator of hydroxide formation within the layered structure. Interestingly, carbon content also rises, likely due to the increased surface area and nanostructured morphology that can trap ambient carbon-containing species. These changes correlate well with SEM observations, where a flake-like, partially porous coating was seen covering the mesh structure. The Ni and Co content balanced with oxygen suggests the formation of a catalytically active NiCo-LDH phase, ideal for enhancing the hydrogen evolution reaction (HER). The nanosheet morphology observed enhances the electrochemically active surface area, while the Ni-Co synergy is known to facilitate improved proton reduction kinetics.

In the case of sample (c), the NiFe-LDH/SSM, the atomic composition further validates the formation of a different LDH architecture. Fe and Ni atoms are both present in significant proportions, with Fe content surpassing that in the bare substrate, indicating deliberate integration rather than substrate exposure. The oxygen atomic % is even higher than in the NiCo sample, which suggests a more developed or thicker hydroxide network. This corresponds directly to the SEM results, where a dense, flower-like microstructure was observed, indicating extensive deposition and surface coverage. Carbon content remains elevated, possibly due to the high surface area and the intrinsic porosity of the coating. Notably, Co is absent, confirming the distinct composition of this NiFe system.

These compositional profiles are essential for interpreting the catalytic roles of each electrode. In NiFe-LDH/SSM, the higher oxygen content and balanced Ni-Fe ratio imply a favorable structure for the urea oxidation reaction (UOR), as Fe sites are known to facilitate urea adsorption and oxidative dehydrogenation steps. The enhanced surface coverage, supported by both SEM and oxygen enrichment in EDX, implies more active catalytic interfaces and better electron transfer pathways. Meanwhile, the NiCo-LDH/SSM configuration is optimized for HER, benefiting from the synergistic catalytic activity of Ni and Co in basic media.

In summary, EDX atomic % analysis confirms the successful and compositionally distinct formation of NiCo-LDH and NiFe-LDH coatings on stainless steel mesh. These results, in strong agreement with the corresponding SEM data, highlight the deliberate compositional tuning of the electrode surface for application-specific catalytic performance in urea electrolysis. The integration of Ni-Co and Ni-Fe with oxygen-rich phases illustrates the tailored design of electrocatalysts for industrial and medical wastewater treatment.

Conclusion

The superior performance of NiFe-LDH in the urea oxidation reaction (UOR) is attributed to the synergistic interaction between Ni^{2+}/Ni^{3+} and Fe^{3+} species, which enhances electron conductivity and promotes the formation of high-valent Ni^{3+}-based active sites, crucial for urea cleavage. The improved conductivity, charge separation, and the layered structure of LDHs facilitate urea diffusion, thus further enhancing the electrocatalytic activity. Among other materials, NiCu-LDH/SSM demonstrated favorable UOR performance, though its onset potential and current density were lower than those of NiCd-LDH and NiCo-LDH exhibited relatively lower UOR performance, likely due to limited redox-active species or ineffective redox transitions for urea oxidation. FTIR analysis confirmed the successful formation of the NiFe-LDH structure, including intercalated nitrate anions and hydroxyl groups, which are essential for maintaining the integrity of the layered architecture. These structural features support the potential of NiFe-LDH for electrocatalytic applications, such as hydrogen evolution reactions (HER) and UOR, due to the synergistic interactions between nickel and iron centers.

Moreover, first-principles calculations based on density functional theory (DFT) provided more profound insights into the electronic structure, highlighting the favorable electrochemical properties of NiFe-LDH. This combination of experimental and theoretical approaches underscores the promising potential of NiFe-LDH composites in various electrocatalytic applications, including direct urea fuel cells and wastewater treatment.

References

Abdelfattah, S. A., Omran, M. M., Mohamed, A. et al. 2025. In-situ preparation of NiCo-LDH/NF electrode for green hydrogen production through alkaline water electrolysis. *Materials Chemistry and Physics* **332**: 130097.

Abraham, D. S., Vinoba, M., and Bhagiyalakshmi, M. 2025. *10.1039/D4NR04064C*. NiCr-LDH/V(4)C(3) MXene nanocomposites as an efficient electrocatalyst for urea oxidation. *Nanoscale* **17** (7): 4111-4122.

Babar, P., Lokhande, A., Karade, V. et al. 2019. Trifunctional layered electrodeposited nickel iron hydroxide electrocatalyst with enhanced performance towards the oxidation of water, urea and hydrazine. *Journal of Colloid and Interface Science* **557**: 10-17.

Babar, P., Lokhande, A., Karade, V. et al. 2019. Bifunctional 2D Electrocatalysts of Transition Metal Hydroxide Nanosheet Arrays for Water Splitting and Urea Electrolysis. *Acs Sustainable Chemistry & Engineering* **7** (11): 10035-10043.

Cai, X. Q., Shen, X. P., Ma, L. B. et al. 2015. Solvothermal synthesis of NiCo-layered double hydroxide nanosheets decorated on RGO sheets for high performance supercapacitor. *Chem Eng J* **268**: 251-259.

Cai, Z. Y., Bu, X. M., Wang, P. et al. 2019. Recent advances in layered double hydroxide electrocatalysts for the oxygen evolution reaction. *Journal of Materials Chemistry A* **7** (10): 5069-5089.

Chen, B., Humayun, M., Li, Y. et al. 2021. Constructing Hierarchical Fluffy CoO–Co4N@NiFe-LDH Nanorod Arrays for Highly Effective Overall Water Splitting and Urea Electrolysis. *ACS Sustainable Chemistry & Engineering* **9** (42): 14180-14192.

Chen, G. F., Ma, T. Y., Liu, Z. Q. et al. 2016. Efficient and Stable Bifunctional Electrocatalysts Ni/Ni_xM_y (M = P, S) for Overall Water Splitting. *Advanced Functional Materials* **26** (19): 3314-3323.

Chen, R., Hung, S. F., Zhou, D. et al. 2019. Layered Structure Causes Bulk NiFe Layered Double Hydroxide Unstable in Alkaline Oxygen Evolution Reaction. *Adv Mater* **31** (41): e1903909.

Chen, S., Duan, J. J., Vasileff, A. et al. 2016. Size Fractionation of Two-Dimensional Sub-Nanometer Thin Manganese Dioxide Crystals towards Superior Urea Electrocatalytic Conversion. *Angewandte Chemie-International Edition* **55** (11): 3804-3808.

Chen, Y., Wu, Y., Chen, M. et al. 2025. Interface engineering of Ni3S2@NiFe-LDH core-shell heterostructure to achieve energy-saving hydrogen production. *Journal of Alloys and Compounds* **1010**: 177855.

Chun Yin, F. Y., Shuli Wang, Ligang Feng. 2023. Heterostructured NiSe2/MoSe2 electronic modulation for efficient electrocatalysis in urea assisted water splitting reaction. *Chinese Journal of Catalysis* **51**: 225-236.

Clark, S. J., Segall, M. D., Pickard, C. J. et al. 2005. First principles methods using CASTEP. *Z Kristallogr* **220** (5-6): 567-570.

Fan, K., Chen, H., Ji, Y. F. et al. 2016. Nickel-vanadium monolayer double hydroxide for efficient electrochemical water oxidation. *Nature Communications* **7**: 1-9.

Fan, R. L., Mu, Q. Q., We, Z. H. et al. 2020. Atomic Ir-doped NiCo layered double hydroxide as a bifunctional electrocatalyst for highly efficient and durable water splitting. *Journal of Materials Chemistry A* **8** (19): 9871-9881.

Gao, R., Zhang, H., and Yan, D. P. 2017. Iron diselenide nanoplatelets: Stable and efficient water-electrolysis catalysts. *Nano Energy* **31**: 90-95.

Gong, Y. M., Zhao, H. B., Ye, D. X. et al. 2022. High efficiency UOR electrocatalyst based on crossed nanosheet structured FeCo-LDH for hydrogen production. *Appl Catal a-Gen* **643**: 118745.

Guo, Z. G., Ye, W., Fang, X. Y. et al. 2019. Amorphous cobalt-iron hydroxides as high-efficiency oxygen-evolution catalysts based on a facile electrospinning process. *Inorganic Chemistry Frontiers* **6** (3): 687-693.

Hu, C. L., Zhang, L., and Gong, J. L. 2019. Recent progress made in the mechanism comprehension and design of electrocatalysts for alkaline water splitting. *Energy & Environmental Science* **12** (9): 2620-2645.

Huang, C., Pi, C. R., Zhang, X. M. et al. 2018. In Situ Synthesis of MoP Nanoflakes Intercalated N-Doped Graphene Nanobelts from MoO_3-Amine Hybrid for High-Efficient Hydrogen Evolution Reaction. *Small* **14** (25): 1800667.

Jadhav, H. S., Roy, A., Desalegan, B. Z. et al. 2020. An advanced and highly efficient Ce assisted NiFe-LDH electrocatalyst for overall water splitting. *Sustainable Energy & Fuels* **4** (1): 312-323.

Kong, D., Zheng, Y., Kobielusz, M. et al. 2018. Recent advances in visible light-driven water oxidation and reduction in suspension systems. *Mater Today* **21** (8): 897-924.

L.M. Le Leuch, T. J. B. 2007. The role of water and surface acidity on the reactive adsorption of ammonia on modified activated carbons. *Carbon* **45** (3): 568-578.

Li, J. X., Sun, S. J., Yang, Y. et al. 2022. An efficient heterogeneous Ni/Ni_2P catalyst for urea-assisted water electrolysis. *Chemical Communications* **58** (68): 9552-9555.

Li, J. X., Yin, C., Wang, S. L. et al. 2024. Built-in electrophilic/nucleophilic domain of nitrogen-doped carbon nanofiber-confined Ni_2P/Ni_3N nanoparticles for efficient urea-containing water-splitting reactions. *Chemical Science* **15** (34): 13659-13667.

Li, P., Duan, X., Kuang, Y. et al. 2018. Tuning Electronic Structure of NiFe Layered Double Hydroxides with Vanadium Doping toward High Efficient Electrocatalytic Water Oxidation. *Advanced Energy Materials* **8** (15): 1703341.

Li, P., Li, W., Huang, Y. et al. 2023. Unconventional Phase Synergies with Doping Engineering Over Ni Electrocatalyst Featuring Regulated Electronic State for Accelerated Urea Oxidation. *ChemSusChem* **16** (6): e202201921.

Li, Y., Wei, X., Chen, L. et al. 2021. Electrocatalytic Hydrogen Production Trilogy. *Angew Chem Int Ed Engl* **60** (36): 19550-19571.

Liao, H. B., Sun, Y. M., Dai, C. C. et al. 2018. An electron deficiency strategy for enhancing hydrogen evolution on CoP nano-electrocatalysts. *Nano Energy* **50**: 273-280.

Long, G., Chen, L., Chen, X. et al. 2025. Interface engineering in Ni(OH)(2)/NiOOH heterojunction to enhance energy-efficient hydrogen production via urea electrolysis. *J Colloid Interface Sci* **680** (Pt A): 880-889.

Long, X., Wang, Z. L., Xiao, S. et al. 2016. Transition metal based layered double hydroxides tailored for energy conversion and storage. *Mater Today* **19** (4): 213-226.

Lotfi, N., Shahrabi, T., Yaghoubinezhad, Y. et al. 2019. Electrodeposition of cedar leaf-like graphene Oxide@Ni-Cu@Ni foam electrode as a highly efficient and ultra-stable catalyst for hydrogen evolution reaction. *Electrochim Acta* **326**: 134949.

Lv, Q. L., Guo, H. R., Xu, D. et al. 2025. Interfacial engineering of NiMn-LDH@Rh(OH) Heterojunctions for Promoted electrocatalytic urea oxidation. *Applied Surface Science* **700**: 163158.

Mengya Yang, R. Z., Yi Liu, Hua Lin. 2022. LDH-derived phosphide/N-doped graphene oxide hierarchical electrocatalyst for enhanced oxygen evolution reaction. *Crystengcomm* **24**: 1189-1194.

Perdew, J. P., Burke, K., and Wang, Y. 1996. Generalized gradient approximation for the exchange-correlation hole of a many-electron system. *Physical Review B* **54** (23): 16533-16539.

Pirkarami, A., Rasouli, S., and Ghasemi, E. 2019. 3-D CdS@NiCo layered double hydroxide core-shell photoelectrocatalyst used for efficient overall water splitting. *Appl Catal B-Environ* **241**: 28-40.

Qin, K. Q., Wang, L. P., Wen, S. W. et al. 2018. Designed synthesis of NiCo-LDH and derived sulfide on heteroatomdoped edge-enriched 3D rivet graphene films for high-performance asymmetric supercapacitor and efficient OER. *Journal of Materials Chemistry A* **6** (17): 8109-8119.

Sahoo, D. P., Das, K. K., Mansingh, S. et al. 2022. Recent progress in first row transition metal Layered double hydroxide (LDH) based electrocatalysts towards water splitting: A review with insights on synthesis. *Coord Chem Rev* **469**: 214666.

Satheesan, A. K., Madhu, R., Nagappan, S. et al. 2025. *10.1039/D4CC05405A*. Current progress in layered double hydroxide-based electrocatalysts for urea oxidation: insights into strategies and mechanisms. *Chem Commun (Camb)* **61** (21): 4092-4109.

Shenyi Song, X. H., Yun Yang, Ligang Feng. 2024. Se self-doped Ni(OH)2 for an efficient urea oxidation reaction. *Chemical Communications* **60**: 10906-10909.

Song, F. and Hu, X. L. 2014. Ultrathin Cobalt-Manganese Layered Double Hydroxide Is an Efficient Oxygen Evolution Catalyst. *Journal of the American Chemical Society* **136** (47): 16481-16484.

Sun, H. C. A., Zhang, W., Li, J. G. et al. 2021. Rh-engineered ultrathin NiFe-LDH nanosheets enable highly-efficient overall water splitting and urea electrolysis. *Appl Catal B-Environ* **284**: 119740.

Sun, X. J. and Ding, R. 2020. Recent progress with electrocatalysts for urea electrolysis in alkaline media for energysaving hydrogen production. *Catalysis Science & Technology* **10** (6): 1567-1581.

Sung Dae Yim, S. J. K., Joon Hyun Baik, In−Sik Nam, Young Sun Mok, Jong-Hwan Lee, Byong K. Cho, Se H. Oh. 2004. Decomposition of Urea into NH3 for the SCR Process. *Industrial & Engineering Chemistry Research* **43** (16): 4856–4863.

Tang, Y. H., Liu, Q., Dong, L. et al. 2020. Activating the hydrogen evolution and overall water splitting performance of NiFe LDH by cation doping and plasma reduction. *Appl Catal B-Environ* **266**: 118627.

Teng, Y., Wang, X. D., Chen, H. Y. et al. 2017. Iron-assisted engineering of molybdenum phosphide nanowires on carbon cloth for efficient hydrogen evolution in a wide pH range. *Journal of Materials Chemistry A* **5** (43): 22790-22796.

Wang, C., Lu, H. L., Mao, Z. Y. et al. 2020. Bimetal Schottky Heterojunction Boosting Energy-Saving Hydrogen Production from Alkaline Water via Urea Electrocatalysis. *Advanced Functional Materials* **30** (21): 2000556.

Wang, X. D., Cao, Y., Teng, Y. et al. 2017. Large-Area Synthesis of a Ni(2)P Honeycomb Electrode for Highly Efficient Water Splitting. *ACS Appl Mater Interfaces* **9** (38): 32812-32819.

Wang, Y., Yan, D., El Hankari, S. et al. 2018. Recent Progress on Layered Double Hydroxides and Their Derivatives for Electrocatalytic Water Splitting. *Adv Sci (Weinh)* **5** (8): 1800064.

Wang, Z., Zhou, Q. Y., Zhu, Y. N. et al. 2022. NiFeMn-Layered Double Hydroxides Linked by Graphene as High-Performance Electrocatalysts for Oxygen Evolution Reaction. *Nanomaterials* **12** (13): 2200.

Wu, Z., Hu, H., Zhang, H. et al. 2025. Enhancement of the urea oxidation reaction by constructing hierarchical CoFe-PBA@S/NiFe-LDH nanoboxes with strengthened built-in electric fields. *Journal of Colloid and Interface Science* **682**: 324-331.

Yan, L., Sun, Y. L., Hu, E. L. et al. 2019. Facile in-situ growth of Ni_2P/Fe_2P nanohybrids on Ni foam for highly efficient urea electrolysis. *Journal of Colloid and Interface Science* **541**: 279-286.

Yan, W., Wang, D., and Botte, G. G. 2012. Electrochemical decomposition of urea with Ni-based catalysts. *Appl Catal B-Environ* **127**: 221-226.

Yan, Y., Shi, X. R., Miao, M. et al. 2018. Bio-inspired design of hierarchical FeP nanostructure arrays for the hydrogen evolution reaction. *Nano Research* **11** (7): 3537-3547.

Yang, M., Zhao, R., Liu, Y. et al. 2022. *10.1039/D1CE01621K*. LDH-derived phosphide/N-doped graphene oxide hierarchical electrocatalyst for enhanced oxygen evolution reaction. *Crystengcomm* **24** (6): 1189-1194.

Yang, W. L., Yang, X. P., Hou, C. M. et al. 2019. Rapid room-temperature fabrication of ultrathin Ni(OH) nanoflakes with abundant edge sites for efficient urea oxidation. *Appl Catal B-Environ* **259**: 118020.

Yang, W. L., Yang, X. P., Li, B. J. et al. 2019. Ultrathin nickel hydroxide nanosheets with a porous structure for efficient electrocatalytic urea oxidation. *Journal of Materials Chemistry A* **7** (46): 26364-26370.

Yang, Y. Q., Zhang, W. B., Xiao, Y. L. et al. 2019. CoNiSe heteronanorods decorated with layered-double-hydroxides for efficient hydrogen evolution. *Appl Catal B-Environ* **242**: 132-139.

Ye, W., Fang, X. Y., Chen, X. B. et al. 2018. A three-dimensional nickel-chromium layered double hydroxide micro/nanosheet array as an efficient and stable bifunctional electrocatalyst for overall water splitting. *Nanoscale* **10** (41): 19484-19491.

Yuanjian Li, H. W., Rui Wang, Beibei He, Yansheng Gong. 2018. 3D self-supported Fesingle bondOsingle bondP film on nickel foam as a highly active bifunctional electrocatalyst for urea-assisted overall water splitting. *Materials Research Bulletin* **100**: 72-75.

Zeng, Z. Y., Chen, Y., Yang, X. J. et al. 2025. Role of Photorelaxation effect in stability and activity of K doped TiO2/NiFe-LDH Composites for oxygen evolution reaction. *Renewable Energy* **248**: 123078.

Zhang, H. J., Maijenburg, A. W., Li, X. P. et al. 2020. Bifunctional Heterostructured Transition Metal Phosphides for Efficient Electrochemical Water Splitting. *Advanced Functional Materials* **30** (34): 2003261.